

Katelyn (Kae) Chi

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Education

Boston University

Expected in 12/2026

Majors: Computer Science and Philosophy

Work Experience

Software Development Engineer Intern

06/2026 – 09/2026

Amazon Web Services – Seattle, Washington

- Built OS inference module for internal AWS AMI fleet, enabling vulnerability assessment for tens of millions of AMIs across 39 distinct regions.
- Implemented lineage-based OS inference algorithm, achieving 389× the throughput of the ported name-based inference logic while increasing OS classification coverage from 56% to 100% with full accuracy fleet-wide.
- Migrated name-based OS inference logic from AWS Glue job to Rust-based AWS Fargate inventory job, achieving 2.5× single-core classification throughput, enabling the deprecation of redundant infrastructure and a separate on-call rotation.
- Productionized the system using AWS CDK, AWS AppConfig feature flags, failure isolation, and 30+ CloudWatch EMF metrics backed by operational alarms; validated with unit, property-based, and end-to-end tests.

Software Engineer

09/2025 – 05/2026

Fidelity Investments – Boston, Massachusetts

- Built the internal company hardware provisioning platform, the life cycle management tool for 20,000+ production machines through a GUI and deployable API supporting compute, storage, and networking resources as a full time student.
- Implemented full lifecycle management for Oracle virtual machines in Go, building API endpoints for create, resize, and delete operations enabling internal teams to provision, reconfigure, and automate deployment of virtual machines.
- Developed 40+ unit and integration tests in Go using httpmock and testify/assert, ensuring robust RESTful API validation and driving test coverage above 90%.

Software Engineering Intern

06/2025 – 08/2025

Fidelity Investments – Boston, Massachusetts

- Led design, development, and production of an internal observability microservice in collaboration with 5 business units and 3 teams; built reusable Go APIs to aggregate Datadog data into a unified JSON endpoint, deployed using a Kubernetes cron job, enabling more metrics for real-time dashboard updates.
- Implemented RESTful APIs in Go to expose 30+ system metrics and automate Datadog monitor creation in a Linux environment, enabling service level objective tracking and automated error threshold detection.

Software Development Co-op

01/2025 – 05/2025

Nokia – Naperville, Illinois

- Developed mission-critical GUI for the 5G Banshee Flex Radio, a portable, self-contained LTE/5G network supporting 1,000 concurrent users.
- Built and deployed 20+ RESTful APIs using Python and FastAPI, supporting multi-platform compatibility, executing efficient SQL queries, interfacing with network switches, DHCP servers, custom hardware modules, and multithreaded polling processes.

Skills

Languages: Python, Go, Rust, Java, JavaScript/Typescript, C, C++, HTML/CSS, SQL.
Frameworks/Databases: React, FastAPI, Flask, Echo, Ginkgo, Node.js, DynamoDB, PostgreSQL, SQLite, Springboot.
Cloud & Infra: AWS (EC2, S3, Lambda, CloudWatch, CDK/CloudFormation, Step Functions), Kubernetes, Docker, Jenkins.
Development Tools: Git, Docker, Linux, Kubernetes, Jira, Wireshark, GDB, Sonarqube.